Predicting Movie Ratings — MovieLens 10M

HarvardX Data Science Capstone Project

Kouramoudou MAGASSOUBA

October 26, 2026

Contents

1 Executive Summary **1**

2 Introduction **2**

3 Data and Evaluation Protocol **2**

4 Exploratory Data Analysis **2**

4.1 Rating Distribution 2

4.2 Movies, Genres, and Activity 3

5 Methodology **5**

5.1 Evaluation Metric 5

5.2 Candidate Models 5

6 Results **6**

6.1 Validation Model Comparison 6

6.2 Regularization Tuning 6

6.3 Independent Final Holdout Evaluation 7

7 Discussion **8**

8 Conclusion **8**

9 Reproducibility and References **8**

1 Executive Summary

This project develops a movie-rating prediction system using the MovieLens 10M dataset. Five model configurations are compared on a validation subset drawn exclusively from `edx`. The final holdout set is not used for model development or hyperparameter selection. The selected model is **Regularized + exact genre combinations**, with regularization parameter $\lambda = 5.0$. Its validation RMSE is **0.864942**, and its final RMSE on `final_holdout_test` is **0.864450**.

2 Introduction

A movie recommendation system seeks to estimate how a user will rate a movie. The aim is to minimize prediction error while maintaining an independent, reproducible evaluation procedure. We examine a global-mean baseline, movie and user bias models, regularization, and an additional effect for exact genre combinations.

3 Data and Evaluation Protocol

MovieLens 10M contains 10,000,054 ratings. The project split produced **9,000,055** observations in `edx` and **999,999** in `final_holdout_test`. The internal development split contains **7,200,089** training observations and **1,799,966** validation observations.

The variables are `userId`, `movieId`, `rating`, `timestamp`, `title`, and `genres`. Every movie and user in the final holdout also appears in `edx`, following the supplied course preparation procedure.

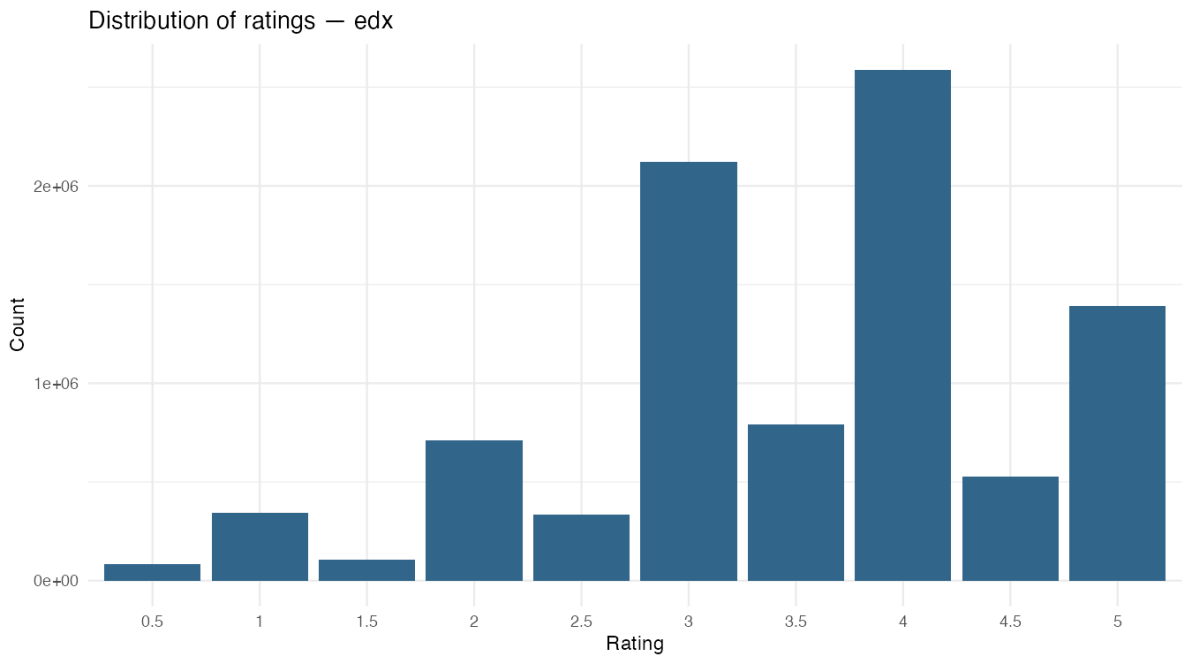
Strict separation: all model and hyperparameter choices are based on the internal validation set. The final holdout is evaluated once, after the selected model is refitted on all of `edx`.

4 Exploratory Data Analysis

4.1 Rating Distribution

Table 1: Rating distribution in edx

Rating	Number of ratings
0.5	85374
1.0	345679
1.5	106426
2.0	711422
2.5	333010
3.0	2121240
3.5	791624
4.0	2588430
4.5	526736
5.0	1390114



4.2 Movies, Genres, and Activity

Table 2: Ten most frequently rated movies

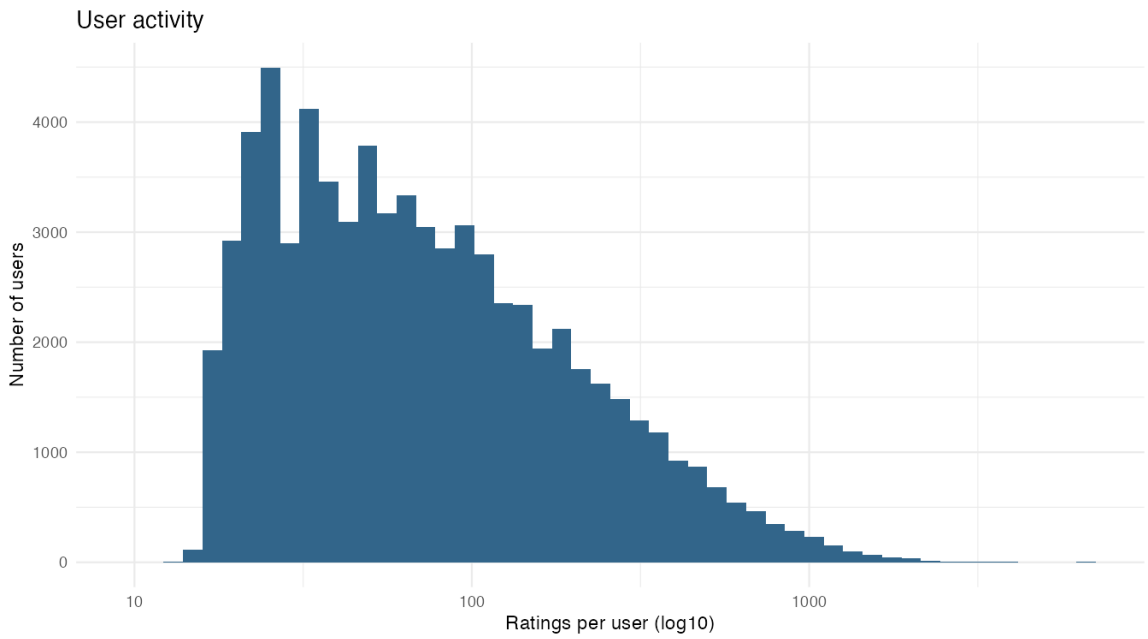
Movie ID	Title	Number of ratings
296	Pulp Fiction (1994)	31362
356	Forrest Gump (1994)	31079
593	Silence of the Lambs, The (1991)	30382
480	Jurassic Park (1993)	29360
318	Shawshank Redemption, The (1994)	28015
110	Braveheart (1995)	26212
457	Fugitive, The (1993)	25998
589	Terminator 2: Judgment Day (1991)	25984
260	Star Wars: Episode IV - A New Hope (a.k.a. Star Wars) (1977)	25672
150	Apollo 13 (1995)	24284

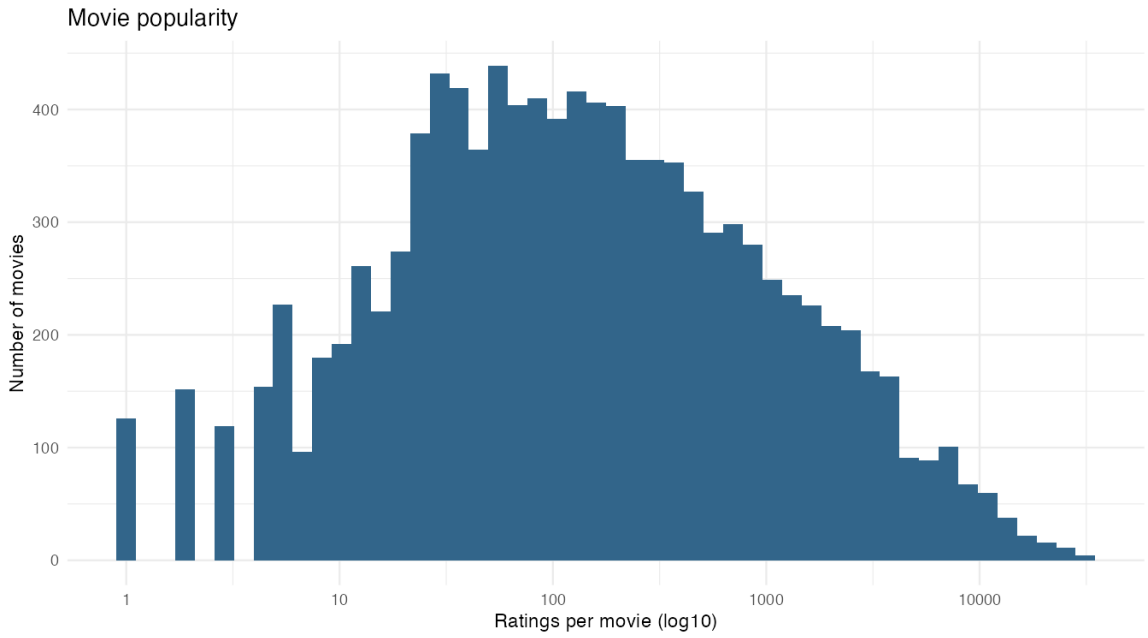
Table 3: Selected genre frequencies

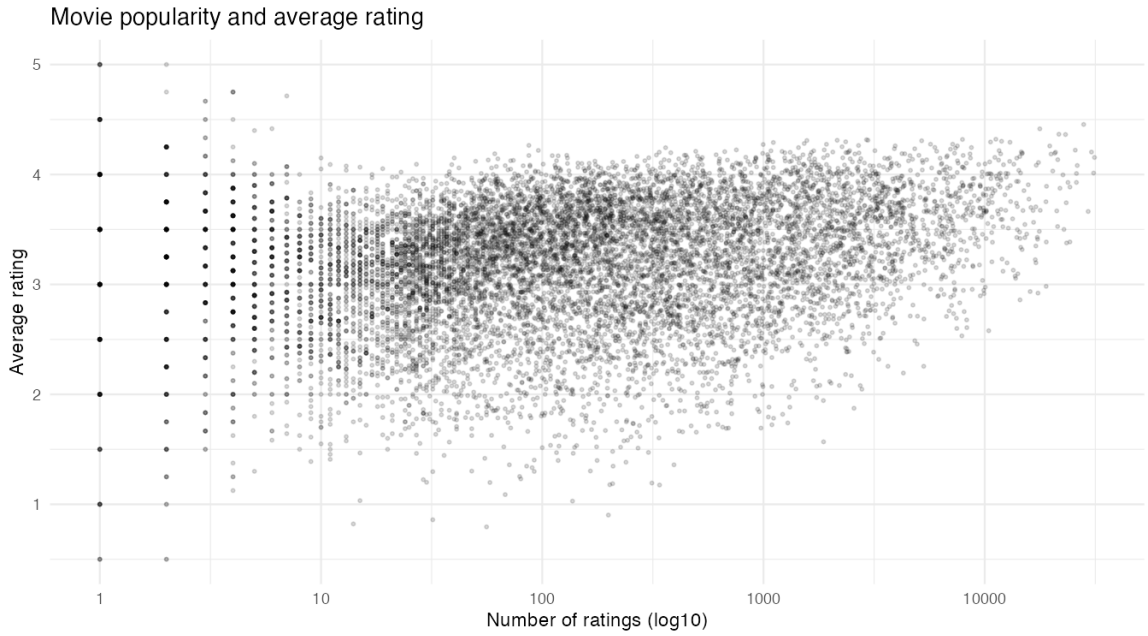
Genre	Number of ratings
Drama	3910127
Comedy	3540930
Thriller	2325899
Romance	1712100

The most frequent rating in `edx` is **4**, and **50.06%** of ratings are at least 4. This provides useful context for interpreting the global-mean baseline and the distribution of prediction errors.

Genres are not mutually exclusive: a movie may belong to multiple categories. The genre model uses each movie's **exact genre combination**, not separate coefficients for individual genres.


User activity
Number of users
4000
3000
2000
1000
0
10
100
1000
Ratings per user (log10)


Movie popularity
Number of movies
400
300
200
100
0
1
10
100
1000
10000
Ratings per movie (log10)



5 Methodology

5.1 Evaluation Metric

Root mean squared error (RMSE) is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(r_j - \hat{r}_j)^2}.$$

Lower values indicate better predictive accuracy.

5.2 Candidate Models

The first model predicts the global mean $\hat{r}_{ui} = \mu$. The second adds a movie-specific bias b_i, and the third also includes a user-specific bias b_u:

$$\hat{r}_{ui} = \mu + b_i + b_u.$$

Regularization shrinks effects estimated from relatively few observations. For example,

$$b_i = \frac{\sum_u (r_{ui} - \mu)}{n_i + \lambda}, \qquad b_u = \frac{\sum_i (r_{ui} - \mu - b_i)}{n_u + \lambda}.$$

The fifth configuration adds a bias b_g for the movie's exact genre combination:

$$\hat{r}_{ui} = \mu + b_i + b_u + b_g.$$

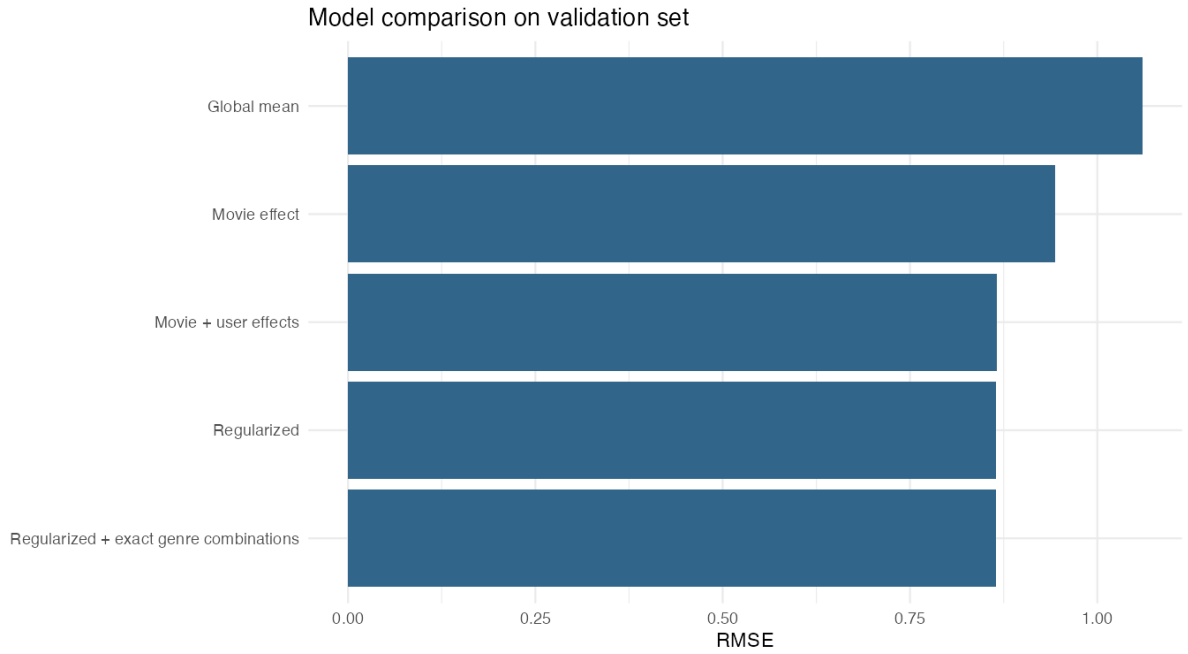
Biases are fitted **sequentially**, rather than by joint optimization. The regularization grid runs from 0 to 10 in increments of 0.5. Both the model family and the value of λ are selected using validation RMSE only.

6 Results

6.1 Validation Model Comparison

Table 4: Model comparison on the internal validation set

Model	Prediction type	Lambda	Validation RMSE
Regularized + exact genre combinations	movie_user_genres	5	0.864942
Regularized	movie_user	5	0.865242
Movie + user effects	movie_user	0	0.865932
Movie effect	movie	0	0.943743
Global mean	global	NA	1.059904



The selected model improves validation RMSE over the global-mean baseline by **0.194963** points, equivalent to **18.39%**. This comparison uses the same internal validation observations and does not rely on the final holdout.

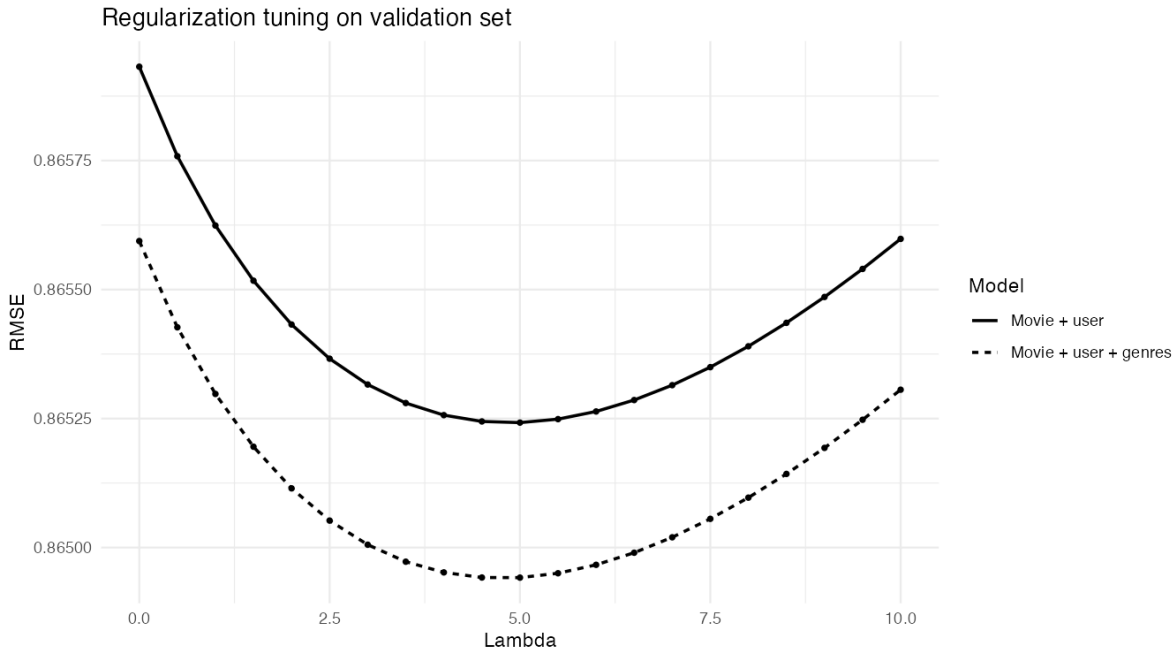
The change in validation RMSE from the unregularized movie-and-user model to its regularized version is **0.000690** points (positive means regularization helped). The difference between the independently tuned regularized model without genres and the version with exact genre combinations is **0.000300** points (positive values favor the genre-augmented model). These are predictive comparisons, not evidence of causal effects.

6.2 Regularization Tuning

Table 5: Regularization grid evaluated on the validation set

Lambda	Regularized RMSE	Genre-augmented RMSE
0.0	0.865932	0.865594

Lambda	Regularized RMSE	Genre-augmented RMSE
0.5	0.865758	0.865427
1.0	0.865624	0.865298
1.5	0.865517	0.865195
2.0	0.865432	0.865115
2.5	0.865366	0.865052
3.0	0.865316	0.865006
3.5	0.865280	0.864973
4.0	0.865257	0.864952
4.5	0.865244	0.864942
5.0	0.865242	0.864942
5.5	0.865249	0.864950
6.0	0.865264	0.864967
6.5	0.865286	0.864990
7.0	0.865315	0.865020
7.5	0.865350	0.865056
8.0	0.865390	0.865097
8.5	0.865436	0.865143
9.0	0.865486	0.865193
9.5	0.865540	0.865248
10.0	0.865598	0.865306



6.3 Independent Final Holdout Evaluation

After the validation-based selection was finalized, **Regularized + exact genre combinations** was refitted on all of `edx` with $\lambda = 5.0$ and applied once to `final_holdout_test`. The resulting RMSE is **0.864450**. The final-minus-validation RMSE difference is **-0.000492** points. This difference is reported descriptively and was not used to revise the model or hyperparameter choices.

7 Discussion

The internal validation procedure makes it possible to assess movie and user effects, regularization, and genre-combination effects without consulting the final holdout. The winning model is **Regularized + exact genre combinations**, with regularization parameter **5.0**. Small differences in RMSE should be interpreted cautiously because they depend on the particular internal split.

Movie and user biases account for systematic variation in average movie ratings and users' rating tendencies. Regularization reduces unstable estimates for sparsely rated movies or less active users. The exact genre-combination effect adds another potential predictor, but its contribution should be judged against simpler models on the same validation set.

Predictions are not clipped to the observed rating range of 0.5 to 5. The same prediction rule is used for validation and final testing. The genre variable represents a complete combination of genres, not separate independent genre effects.

Limitations include the absence of latent factors, detailed movie content features, and temporal dynamics. A single internal split is used; the analysis does not provide confidence intervals for the differences in RMSE.

8 Conclusion

The project demonstrates a complete workflow: data preparation, exploratory analysis, candidate model fitting, validation-based hyperparameter selection, refitting on `edx`, and a reserved final evaluation. The final holdout RMSE is **0.864450**. The generated CSV files, charts, and session metadata support review and reproduction of the results.

9 Reproducibility and References

Run the following command from the directory containing both project files:

```
source("MovieLens_Project.R", encoding = "UTF-8")
```

The script saves model outputs and renders this R Markdown report to `MovieLens_Report.pdf`. A functioning LaTeX distribution (such as TinyTeX or MacTeX) is required.

Reproducibility notes: The script creates a durable marker before final holdout evaluation to guard against accidental repeated testing. If the PDF compilation fails after the model finishes, render this Rmd directly; do not rerun the model script. R and package versions are recorded in `resultats_movielens/session_info.txt`. The `.rds` datasets and intermediate outputs are not part of the three required submission files. The PDF must be produced after actual execution; no RMSE value is hard-coded.

References: GroupLens Research, *MovieLens 10M Dataset*; HarvardX, *Data Science Professional Certificate*, MovieLens project; R documentation for `caret`, `dplyr`, `ggplot2`, `knitr`, and `rmarkdown`.